



Sheryians  
Coding School

# POWER BI

Master Data Cleaning, Transformation, And Loading Using  
Power Query.

P H A S E - 2

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Power BI

## GET DATA

### Identify & Connect To Data Source

Before Loading Data, A Data Analyst Must:

- Understand Where Data Is Coming From
- Identify File Type, Size, And Structure
- Check Refresh Requirement

Common Sources:

- Excel / CSV
- SQL Server
- Web APIs
- SharePoint
- Folder (Multiple Files)

Correct Source Selection Improves Performance & Reliability.



## Select A Storage Mode

Storage Mode Defines How Power BI Stores & Queries Data.

### Import Mode

- Storage Mode DData Is Stored In Power BI
- Fast Performance
- Scheduled Refresh Required

#### Best For:

- Small To Medium Datasetefines How Power BI Stores & Queries Data.



## DirectQuery

- Data Stays In Source
- Real-Time Data
- Slower Visuals

Best For:

- Large Databases
- Live Systems

## Dual Mode

- Combination Of Import + DirectQuery
- Smart Caching

Used In:

- Complex Models



## DirectLake

- Used With Microsoft Fabric
- Reads Data Directly From Lake
- Best Performance For Large Data

## Fix Performance Issues

Performance Issues Happen Due To:

- Too Many Columns
- Unused Data
- Wrong Storage Mode
- Complex Transformations

Best Practices:

- Remove Unused Columns Early
- Filter Rows In Power Query
- Use Import When Possible
- Reduce Cardinality



## Resolve Data Import Errors

### Common Errors:

- Column Mismatch
- Data Type Conflict
- Missing Files
- Permission Issues

### Fix Using:

- Power Query Error Handling
- Correct Data Types
- Proper Credentials



# POWER QUERY EDITOR

## Power Query Editor

Used To:

- Clean Data
- Transform Structure
- Apply Business Logic

## M Language Basics

Power Query Uses M Language Behind The Scenes.

Example: `= Table.RemoveRowsWithErrors(Source)`

You Don't Need To Write M At Start, But  
Understanding Helps Optimization.



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## Data Profiling

Used To Check Data Quality.

Tools:

- Column Quality
- Column Distribution
- Column Profile

Shows:

- Null %
- Errors
- Unique Values



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# CLEAN, TRANSFORM & LOAD DATA

## Resolve Data Quality Issues

Data Issues Include:

- Null Values
- Duplicates
- Wrong Formats
- Inconsistent Naming

Fix Using:

- Replace Values
- Remove Duplicates
- Fill Down / Fill Up
- Trim & Clean



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# Data Shape Transformations

## Common Transformations:

- Split Columns
- Merge Columns
- Change Data Types
- Rename Columns
- Reorder Columns

## Goal:

Make Data Analysis-Friendly.



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## User-Friendly Naming Conventions

Good Column Names:

- Short
- Clear
- Business-Friendly

**Bad** : Amt\_2023\_x

**Good** : Total Sales

## Merge Queries Vs Append Queries

**Merge Queries**

- Combine Tables Side-By-Side
- Similar To JOIN

**Example:**

- Sales + Product Table



## Append Queries

- Stack Tables Vertically

Example:

- Jan + Feb + Mar Data

## Conditional & Custom Columns

### Conditional Columns

Used For Logic-Based Rules.

Example:

- If Sales > 50000 → High Else Low

### Custom Columns

Used To Write Expressions.

Example: Sales[Revenue] - Sales[Cost]



# Group By, Aggregate, Pivot & Unpivot

## Group By

Summarizes Data.

Example:

- Total Sales By Region

## Aggregate

Sum, Count, Average, Min, Max

## Pivot Columns

Converts Rows Into Columns.

Example:

- Months As Columns



## Unpivot Columns

Converts Columns Into Rows

Very Common In Real-World Data.

## Handling Errors & Nulls Like A Pro

Techniques:

- Replace Errors With Null
- Remove Error Rows
- Fill Missing Values
- Use Default Values

Goal:

No Broken Visuals Or Wrong Calculations.





# Parameters & Functions In Power Query

## Parameters

Used To:

- Control Values Dynamically
- Change File Paths
- Control Filters

## Functions

Reusable Logic In Power Query.

Example:

- Same Cleaning Steps For Multiple Files





## Dynamic Source Switching

### Folder + File Parameterization

Used When:

- Monthly Files
- Changing Source Location

Process:

- Create Folder Source
- Use Parameter For File Path
- Auto Refresh With New Files

Very Important For Automation.



## Creating Date Table Using M Language

Date Table Is Mandatory For:

- Time Intelligence
- YTD, MTD, YoY Analysis

Example:

- Calendar From 2020–2030
- Year, Month, Quarter Columns



# Query Diagnostics

## Performance Optimization

### Used To:

- Identify Slow Queries
- Optimize Steps
- Reduce Load Time

### Tools:

- Start Diagnostics
- Stop Diagnostics



P O W E R B I

## CASE STUDY

### Sales Data Cleaning + Auto Refresh Setup

#### Problem:

- Multiple Monthly Excel Files
- Inconsistent Columns
- Null Values
- Manual Refresh Needed

#### Solution:

1. Folder Data Import
2. Clean & Standardize Columns
3. Append All Files
4. Create Parameters
5. Set Refresh In Power BI Service



## Result:

- Clean Dataset
- Auto Refresh Enabled
- Zero Manual Work



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